



Course Outline for:

Prepared by:

Tech Team, Encoder Bytes Private Limited

- Web Development:

Course Title: Web Development

Duration: 3 Months

Prerequisite/s:

- Basic Computer Skills
- Logical Thinking and Problem-Solving
- Exposure to Coding

Learning Outcomes:

By the end of this course, the trainees should be able to:

- Gain a solid understanding of how websites function, including both front-end and back-end components, and how they work together to create dynamic online experiences.
- Use HTML, CSS, and JavaScript to design responsive, engaging, and interactive web pages that adapt seamlessly across devices, with a focus on clean, user-friendly interfaces.
- Dive into the world of React.js, learning how to build modern, high-performance web applications using its component-based architecture.
- Understand and apply color theory, design layouts, and visual hierarchy for impactful and aesthetically pleasing compositions.
- Learn the basics of server-side programming by creating a simple Node.js server and connecting it with your front-end applications, expanding your development skills beyond just the browser.
- Learn how to deploy your websites on platforms like GitHub Pages, Netlify, and Heroku, gaining the confidence to showcase your work to clients or employers.
- Build a professional portfolio website, learn how to market yourself as a freelance web developer, and acquire the skills to pitch your services and secure your first clients.

Daily Distribution of Course:

Module	Day	Topics
Module 1: <i>Introduction and Basics of Web Development</i>	Days 1	- Overview of course: Front-end, backend & hosting. - Tools and software setup. - How web works: server, client and HTTP.
	Days 2	- Introduction to HTML - Building basic web page - Common HTML elements: headings, paragraph, links, images. - Semantic tags
	Day 3:	- Forms, tables - Accessibility in HTML
Module 2:	Days 4-5	- Introduction to CSS - Styling with CSS: syntax, selectors, properties - Inline, internal and external CSS. - Colors, fonts, spacing - CSS Box Model: margin, border, padding, content - Introduction to responsive design
	Day 6	- Media Queries and breakpoints - Understanding flex box

		Aligning and distributing
Module 3:	Days 7-8	- Practical exercises with layout - Revision of HTML, CSS - Overview of CSS framework: Bootstrap, Tailwind CSS - Navbar, modals and grid
	Day 9	- Installation of tailwind CSS - Responsive design in tailwind CSS - Exercise: Portfolio Website
Module 4:	Days 10-11	- Introduction to JavaScript - What is JS? Adding JS in to webpage - Variables, data types, operators - Practical exercise: Simple math operation - Conditional statements (if-else, switch) - Loops(for, while, do-while) - Exercise: loop for repetitive tasks: Number/pattern printing
	Day 12	- Functions - Defining and invoking functions - Parameters, arguments and return values

		• Scope(local and global variables) • Exercise: Basic calculator
Module 5:	Days 13-14	• Arrays and Objects • Array methods push, pop, splice • Introduction to objects and their properties • Short revision • Exercise: Todo list
	Day 15	• Introduction to DOM • Selection and modification of elements • Exercise: change styles and content dynamically
Module 6:	Days 16-17	• Event listener (click, mouseover, keydown) • Adding, removing and updating elements dynamically • Introduction to Es6 • Let, const, arrow functions • Template literals and destructuring • What is API? How to fetch data.
	Day 18	• Git and git hub • Introduction to react js • Understanding react

		components and component based architecture • Creating react app • Understanding react folder structure s
Module 7:	Days 19-20	• Introduction to backend • Overview of server-side languages • Setting up simple node js server • Introduction to express js • Building basic routes and handling requests • Introduction to mongo db
	Day 21	• Sending data between front end and backend using forms and APIs • Exercise: simple contact form with backend processing
Module 8:	Days 22-23	• Introduction to Hosting • Overview of hosting platforms • Hosting a static site • Managing environment variables • Freelancing
	Day 24	• Final review and wrap-up